



Machine learning and IoT in healthcare: Recent advancements, challenges & future direction

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ABSTRACT

Background: The integration of Machine Learning and Deep Learning with IoT-enabled devices for real-time health monitoring has significantly revolutionized healthcare. These technologies facilitate the analysis of intricate medical datasets, yielding actionable insights that promote evidence-based clinical decision-making. Although significant advancements have been made, there is still an absence of a thorough synthesis regarding current applications, primary challenges, and prospective research directions. This review aims to synthesize recent applications, identify significant gaps, and propose clear direction for future research.

Methodology: A comprehensive narrative review was performed where a systematic literature search was conducted in PubMed and Scopus for studies published between 2020 and 2025. A total of 300 pertinent papers on ML and IoT's applications in healthcare were selected and analyzed to synthesize technological advancements, trade-offs, practical implications, challenges, and potential directions for future research.

Key findings: Neural network models, such as CNNs and ANNs, along with ensemble methods like Random Forest and XGBoost, often attain predictive accuracies ranging from 85 % to 95 %. Advanced technique, like generative imaging models, reinforcement learning, and transformer-based architectures, improve diagnostics, chronic disease management, robotic-assisted surgery, and predictive analytics, while explainable AI promotes clinical trust. Cloud-edge integration utilizing lightweight machine learning models enables real-time, energy-efficient applications, enhancing diagnosis, decision support, personalization, and cost-effectiveness, notwithstanding current challenges.

Conclusion: To conclude, the integration of ML and IoT is transforming healthcare through enhanced monitoring, improved predictive capabilities, and tailored treatment approaches. Addressing persistent limitations is crucial for fully realizing its potential and directing future research in this evolving field.

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1. Introduction

There is a significant demand for the digitalization of healthcare industries, particularly due to advancements in wireless technologies and electronic equipment. Healthcare 5.0 is the most recent paradigm in healthcare systems, integrating technologies such as Internet of Things (IoT) with machine learning.^{1,2} The prevalence of chronic diseases has significantly risen as a result of global economic transformations. The primary threat to human health arises from chronic diseases. Furthermore, if a disease outbreaks, it results in an inflow of patients to hospitals, straining the ability of medical facilities to provide care for all individuals. The worldwide spreading of COVID-19 has exerted pressure on health care facilities globally. Moreover, numerous outpatient departments encounter the issue of congestion in their waiting areas. Patients visiting hospitals present with many diseases, some necessitating urgent medical intervention. The dilemma is further worse when patients requiring emergency care must endure an extended wait in line. The issue is exacerbated in developing nations with inadequately staffed hospitals. A significant number of patients frequently leave without receiving medical care due to hospital overpopulation. The enhancement and maintenance of health is only achievable through the diagnosis and prompt prevention of ailments. Consequently, it is essential to alleviate the strain on healthcare systems while maintaining the quality and standards of healthcare facilities.³

Machine learning and IoT offer potential solutions to alleviate the pressure on healthcare systems. For example, physicians can readily monitor patients' heart impulses using IoT based healthcare monitoring systems and predict heart diseases with Machine learning, hence facilitating accurate diagnoses.⁴ To provide consistent wireless data transfer, numerous wearable devices have been created. The IoT application enables hospitals to monitor the vital signs of patients with chronic diseases. The IoT has ushered in a new era for the healthcare sector, facilitating proactive connections between experts and patients. ML-driven diagnostics can deliver precise and prompt diagnoses, enhancing treatment outcomes. The IoT is anticipated to further proliferate due to developments in sensor technology, networking, and data analytics. With the passage of time, an increased number of medical equipment will be interconnected Through IoT, facilitating real-time monitoring, remote patient management, and the early identification of health concerns. Real-time data transmission enables prompt action, hence diminishing problems and hospital re-admissions. Although transformative changes on ML-IoT healthcare applications are observed in recent times, there are still some shortcomings in the overall synthesis which needs the attention of researchers.

This paper offers comprehensive insights and synthesis into the latest technological advancements and role of machine learning and IoT in health care applications. Researchers can benefit from acquiring a thorough awareness of the recent trends of machine learning & IoT. Adopting these technologies and concepts can result in superior patient care, optimized healthcare operations, and innovative research that may transform the future of healthcare delivery. Another contribution is addressing the challenges related to the implementation of machine learning and IoT in healthcare. This knowledge can enable the researchers to link their studies with the urgent demands and issues in healthcare. This paper also highlights future directions and research prospects, as it assists researchers in pinpointing pertinent, effective topics of study that correspond with the dynamic requirements of the healthcare sector.

The remainder of the work is structured as follows. Section 2 presents the methodology. Section 3 highlights the State-of-the-Art Developments and case Studies in Healthcare ML-IoT Systems. Section 4 emphasizes on the impact of integrated ML-IoT systems in healthcare applications while addressing its role in improving outcomes in healthcare sectors and key benefits in various applications. Section 5 discusses the result section providing analytical findings derived from the study. Section 6 presents the discussion, outlining the challenges and limitations of this integrated technology, while highlighting future directions and research opportunities for emerging researchers. Section 7 concludes the paper.

2. Materials & method

This review was developed as a narrative review and critical analysis of the literature on machine learning and IoT applications in healthcare. Although it's a narrative review, it has followed a structured process to ensure transparency of the paper. The strategy details are shared below.

2.1. Literature search strategy

Relevant publications were identified mainly via academic databases such as Scopus, PubMed, Web of Science, ScienceDirect, and Google Scholar, ensuring thorough coverage of recent research. The search incorporates combined keywords and phrases including "machine learning," "deep learning," "artificial intelligence," "Internet of Things," "IoMT," "smart healthcare," "healthcare 5.0," "remote patient monitoring," "predictive analytics," "clinical decision support," "wearable sensors," and "personalized medicine." Boolean operators (AND/OR) were used to find intersections among machine learning techniques, Internet of Things technologies, and healthcare applications. The search was limited to articles published within the past five years to incorporate recent and relevant studies. Furthermore, reference lists from significant review papers, systematic reviews, and reports were examined manually to identify additional relevant sources. The literature review encompassed various study types, such as observational studies, clinical trials, case studies, and computational experiments, to encompass a broad range of applications, benefits, challenges, and future directions of ML-IoT in healthcare.

2.2. Inclusion criteria

The following outlines the inclusion criteria:

- Articles from peer-reviewed journals, conference proceedings, and book chapters that focus on ML and the IoT applications, and current trends in healthcare.
- Studies published between 2020 and 2025 will be included to ensure coverage of recent advancements.
- Research articles presenting empirical findings, case studies, or thorough analyses of ML-IoT systems across various healthcare applications
- Research published in English and indexed in Scopus or PubMed.

2.3. Exclusion criteria

The exclusion criteria are given below:

- Articles not directly related to healthcare applications of ML or IoT, such as general AI or IoT studies in other industries.
- Studies without sufficient methodological details or empirical evidence
- Publications older than the past five years.
- Non-English articles or those not indexed in reputable databases (Scopus, PubMed)
- Duplicate studies or redundant reports with overlapping data.

2.4. Scope

This review indicates recent research on the integration of machine learning (ML) and Internet of Things (IoT) technologies in healthcare, emphasizing applications, emerging trends, and practical implementations. The research studies selected were examined to both summarize recent advancements and to identify trade-offs, challenges, and limitations related to ML-IoT systems. The review offers essential insights into the effectiveness and feasibility of ML-IoT solutions by analyzing various approaches, highlighting best practices, and examining real-world applications. Additionally, it highlights gaps in existing research and suggests potential directions for future studies, with the objective of guiding both scholarly inquiry and evidence-based implementation in clinical environments.

3. State-of-the-Art Developments and case Studies in Healthcare ML-IoT systems

Machine Learning (ML) is revolutionizing healthcare sectors by the application of methods including supervised learning, reinforcement learning, and deep learning. Supervised learning is mostly used for diagnostics, while reinforcement learning is important for personalized treatment plans and robotics-assisted surgery. Deep learning is employed in image recognition and natural language processing to analyze medical data and perform predictive analytics. The integration of IoT technology in healthcare, including the use of wearable devices and smart sensors, enhances the real-time collection and analysis of data.^{5–7} Machine Learning (ML) and the Internet of Things (IoT) have significantly advanced healthcare through enhancements in predictive analysis, disease forecasting, and diagnostics, among other areas^{8–19}. Fig. 1 shows some of the applications of ML and IoT in the health-care system.

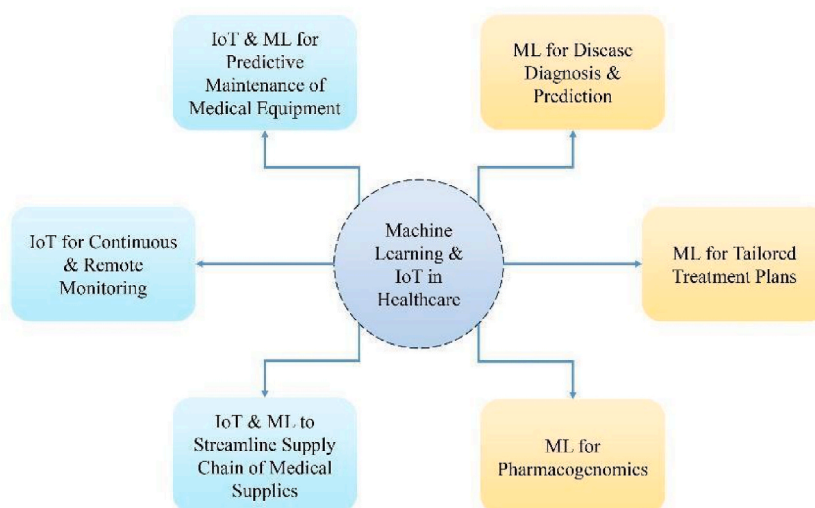


Fig. 1. Applications of ML and IoT in healthcare.

3.1. Machine learning algorithms in healthcare

Machine learning algorithms have improved the capabilities of doctors and analysts by finding complex patterns in medical data. This has made diagnoses more accurate and made it easier to make decisions based on evidence.^{20–28} As a result, these algorithms also make it easier for predictive models in early disease detection, how it will progress, and how to group patients by risk, which helps with personalized treatment plans and proactive healthcare management.^{29–35} Machine learning-based techniques have been applied in large medical centers to increase efficiency in medical imaging and monitoring to identify irregularities in blood samples, organs, bones, robot-assisted surgeries etc..^{36–41} ML makes it possible to analyze the massive amounts of heterogeneous data that are produced every day in the healthcare sector originating from multiple sources. The data include medical records, medical images, data from wearable devices, clinical trial data, data from open sources, genomic data, data from health surveys, and cloud platform. By analyzing such data, ML can uncover patterns, predict disease risks, personalize treatment plans, optimize hospital operations, and support timely clinical decision-making. Fig. 2 illustrates the sources of heterogeneous data analyzed by ML.

As the healthcare industry expands into the modern technological era, numerous innovations are made. The advancement of the discipline depends on methods and applications based on machine learning, which has led to faster, more accurate, and simpler diagnosis & prediction processes.^{42–47} The next subsection outlines the various types of algorithms and their specific applications along with recent trends in healthcare.

3.1.1. Supervised learning for disease diagnosis and prediction

Supervised learning is a machine learning algorithm that uses labeled data to acquire knowledge. The algorithm can be classified into classification and regression algorithms, such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes. These methods are applied when the output variable has a continuous value. Supervised Learning algorithms are used in diagnosis and prediction of different types of diseases including breast cancer, heart diseases, liver disease, thyroid, Alzheimer's disease, lung cancer, prostate cancer, brain cancer, hepatitis, Covid 19, influenza etc.

Breast cancer, the second most diagnosed cancer in women, often relies on mammography, which is only 70 % accurate, leading to potential misdiagnosis. Researchers have applied machine learning for predicting breast cancer by carefully choosing the most effective characteristics, using six supervised learning classifiers: support vector machine, random forest, gradient boosting machine, multilayer perceptron, logistic regression, and K-nearest neighbors (KNN). KNN outperforms all other classifiers on the 20-feature dataset using SVD and achieves a perfect accuracy score of 100 % when employing a PCA with the 15 most significant features, outperforming other classifiers in terms of accuracy.⁴⁸ Other researchers utilized a breast cancer dataset comprising 11 features and implemented 8 machine learning classifiers. The results indicated that Logistic Regression attained the highest testing accuracy of 91.67 % without the application of feature selection. Following the implementation of feature selection, classifiers such as LGBM demonstrated an improvement, achieving an accuracy of 90.74 %.⁴⁹

Cardiovascular diseases result in approximately 17.9 million fatalities each year, with heart attacks and strokes constituting over 80 % of these deaths. Medical researchers face significant challenges in detecting heart disease accurately. A framework has been proposed by researchers that integrates contemporary machine learning techniques to improve heart disease prediction.⁵⁰ Some researchers utilized supervised machine learning classifiers to identify significant features for predicting heart disease. Various classification techniques utilize distinct features within the prediction model. The effectiveness of performance metrics is evaluated in relation to the UCI heart dataset through the application of machine learning techniques, specifically utilizing SMOTE oversampling. The Random Forest method achieved a maximum accuracy of 96.6 %, with sensitivity at 90 %, and both specificity and precision at

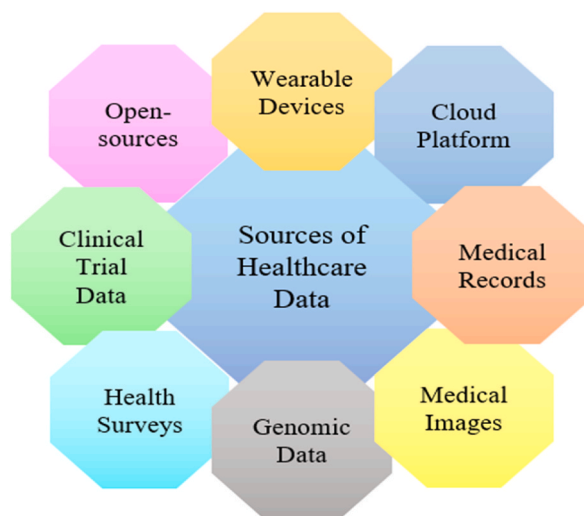


Fig. 2. Different types of heterogeneous data analyzed by ML

100 %.⁵¹ Other researchers applied 13 machine learning methods to predict the occurrence of mortality events. Their model exhibited notable performance on this dataset, attaining an accuracy of 85.23 % with Random Forest across the entire dataset and 86.36 % with Flexible Discriminant Analysis on Extreme Gradient Boosting (XGBoost).⁵²

In the case of liver disease, early detection of liver disease is challenging due to its modest signs and late appearance of symptoms. Five classifiers were employed by some researchers to identify patients with liver disease during the detection phase: Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Naive Bayes (NB), Decision Tree (DT), and Random Forest (RF). The KNN classifier demonstrated the highest classification accuracy among all classifiers evaluated on the test dataset.⁵³ Liver cirrhosis is a disorder that may result from obesity, alcohol intake, viral infections, and autoimmune disorders. A group of researchers analyzed data from 127 patients with alcohol-related liver cirrhosis (ALC). The data were utilized to evaluate the performance of prediction models derived from three machine learning algorithms: K-nearest neighbor (KNN), support vector machine (SVM), and random forest (RF). The SVM-based prediction model demonstrated superior predictive performance compared to alternative models.⁵⁴

Thyroid disease ranks among the most common endocrine disorders globally, highlighting the need for accurate and effective diagnostic models to enhance clinical outcomes. Researchers presented a hybrid Feature Selection and Deep Learning Framework (HFSDLF) that combines Random Forests with Principal Component Analysis (PCA) and L1 regularization. The Random Forest classifier attained the highest accuracy at 96.30 %, surpassing other models including Decision Trees and Logistic Regression, and demonstrating significant enhancements in sensitivity and specificity.⁵⁵

Diabetes mellitus represents a major health challenge due to its association with hyperglycemia. Researchers examined the early detection and management of diabetes through the application of machine learning techniques to electronic health records. A study examined the efficacy of three supervised machine learning algorithms—logistic regression, Random Forest, and k-nearest neighbors (KNN)—in creating predictive models for diabetes. The KNN model was identified as the most effective algorithm for predicting diabetes in the analyzed dataset.⁵⁶ Another study investigated various human body characteristics to predict diabetes. An ensemble method, termed En-RfRsK, is proposed for predicting the risk of diabetes mellitus. This voting classifier integrates three machine learning techniques: random forest (RF), radial support vector machine (R-SVM), and K-nearest neighbor (KNN). The proposed approach demonstrated improved accuracy and predictive capabilities.⁵⁷

Hepatitis is a leading global disease, and machine learning algorithms can help diagnose it by analyzing specific features. Researchers evaluated different classifiers on the UCI data set, including support vector machine, logistic regression, K-nearest neighbor, and random forest. The LR model achieved the highest accuracy of 93.18 % when combined with the SMOTE approach.⁵⁸ Researchers employed machine learning methods to create classification models for hepatitis C, with the objective of predicting the virus responsible for the infection. The employed machine learning algorithms consist of Random Forest, Cat Boost, Bagging Classifier, SGD Classifier, Gaussian Naive Bayes, Bernoulli Naive Bayes, Multinomial Naive Bayes, Linear Discriminant Analysis, Artificial Neural Networks, and Multi-Layer Perceptron. Random forest exhibited superior accuracy among classification methods for diagnosing HCV infection.⁵⁹

Early detection of Alzheimer's disease is crucial for its impact on dementia and the availability of treatments and interventions. Machine learning algorithms like Logistic regression, support vector machine, hierarchical decision tree, ensemble random forest, and boosting AdaBoost are used to detect the disease early, with AdaBoost and random forest algorithms generating more precise results. This early detection allows for better planning and treatment options for Alzheimer's patients.⁶⁰ Other researchers concentrated on implementing various Deep Learning algorithms for the multi-class classification of Alzheimer's disease MRI images. They proposed the most effective pre-trained model capable of accurately predicting the patient's stage. Visualization techniques, including Grad-CAM and Saliency Map, are employed on the model with the highest accuracy to identify the areas of focus within the image that contributed to class prediction.⁶¹

Researchers have been applying machine learning based approaches for early lung cancer diagnosis. A study presents an innovative interpretable hybrid deep learning technique named 'DeepXplainer' designed for lung cancer detection and the elucidation of prediction outcomes. This method utilizes a convolutional neural network in conjunction with XGBoost. The proposed method achieved an accuracy of 97.43 %. Following the model's predictions with a high degree of accuracy, each prediction is elucidated through the application of an explainable artificial intelligence method at both local and global levels.⁶² Other researchers employed a logistic regression-based machine learning model that effectively classifies benign and malignant patients, demonstrating high clinical sensitivity and specificity. Furthermore, significant Raman peak positions associated with various stages of lung cancer were examined, providing insights for the diagnosis of early-stage lung cancer.⁶³

More recently, predictive and diagnostic biomarkers for prostate cancer (PCA) disease's clinical features were found using machine learning algorithms. Researchers developed a Deep Learning-based Prostate Cancer Detection Model (PCDM) for the automated diagnosis of prostate cancer. The PCDM model is an adapted ResNet50 architecture that incorporates faster R-CNN and dual optimizers to enhance detection performance. The proposed model demonstrates elevated sensitivity, specificity, precision, and accuracy rates.⁶⁴ Other researchers' proposed method employs a deep learning framework for the detection of prostate cancer, utilizing the concept of Gleason grading applied to historical images. A 3D convolutional neural network has been employed to analyze the affected region and predict it utilizing the Epithelial and Gleason grading network. The method demonstrates an overall specificity of 85 %, an accuracy of 87 %, and a sensitivity of 89 % at the patient level for various targeted MRI images.⁶⁵

The formation of atypical brain cells, some of which may progress to malignancy, is referred to as a brain tumor. Magnetic Resonance Imaging (MRI) facilitates the segmentation of various regions within brain tumors, a task complicated by variations in location, size, shape, and intensity. Researchers proposed two distinct deep learning models for the detection of both binary (abnormal and normal) and multiclass (glioma, meningioma, and pituitary) brain tumors. A 26-layer Deep CNN, along with fine-tuned VGG16 and Xception models, was developed for the classification of brain tumors from MRI images. Deep CNN attained an accuracy of 97.27

% on the Figshare dataset, while fine-tuned VGG16 and Xception achieved 97.14 % and 98.57 % on the Br35H and Harvard Medical datasets, respectively.⁶⁶ Others proposed a method for detecting brain tumors using an Ensemble Deep Neural Support Vector Machine (EDN-SVM) classifier in MRI scans. The proposed classifier is utilized to classify the abnormal tissues. The numerical results indicate an improved accuracy of 97.93 %, sensitivity of 92 %, and specificity of 98 % in differentiating between aberrant and normal tissue in brain MRI images.⁶⁷

Machine learning algorithms have significantly improved testing and hospital response in the fight against COVID-19, particularly the coronavirus SARS-CoV2. Researchers conducted a study in which various machine-learning algorithms were trained to assess and identify COVID-19-related alterations in lung tissue using X-ray images. X-ray images are categorized into three classifications based on open sources: Patients with COVID-19, those with pneumonia, and healthy individuals were analyzed using existing machine learning and pre-trained deep learning models. Various machine learning models, including K Nearest Neighbor (KNN) and decision trees (DT), alongside transfer learning models such as Convolutional Neural Network (CNN), Visual Geometry Group (VGG-16, VGG-19), ResNet50, DenseNet201, Xception, and InceptionV3, were assessed to evaluate their performance in classifying X-ray images. The comparative analysis demonstrates that VGG-19 with augmentation outperformed the other nine algorithms.⁶⁸ Others developed a simple, dependable, and automated system for assessing the severity of COVID-19 using CT scans, categorizing cases into three stages:

Table 1

Overview of Studies using Supervised Machine Learning for Diagnosis of Different Diseases.

ML Algorithms	Disease Diagnosis	Findings	Ref.
Support vector machine, random forest, gradient boosting machine, multilayer perceptron, logistic regression, and K-nearest neighbors (KNN)	Breast Cancer	KNN surpassed other classifiers using a PCA with the 15 most important features, scoring 100 % accuracy	48
Logistic Regression, Random Forest, Extra Trees, XGB, LGBM, CatBoost, SVC, Gaussian Naïve Bayes		Logistic Regression achieved the highest testing accuracy of 91.67 %	49
Logistic regression, Decision Tree, K-nearest neighbors (KNN), Naïve Bayes (NB) Random Forest (RF), AdaBoost (AB), and Support Vector Machine (SVM)	Heart Disease	Random Forest obtains the most balanced output in terms of precision and specificity of 100 %, along with 90 % sensitivity	51
Logistic regression, Flexible Discriminant Analysis, Support Vector Machine (SVM), Decision Tree, K-nearest neighbors (KNN), Naïve Bayes (NB), Random Forest (RF), XGBoost, Gradient Boosting Machine (GBM), LightGBM, AdaBoost, CatBoost, Neural Networks		A maximum precision of 85.23 % on the entire dataset using and 86.36 % on the XGBoost dataset using Flexible Discriminant Analysis	52
Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Naïve Bayes (NB), Decision Tree (DT), and Random Forest (RF)	Liver Disease	The prediction accuracy using KNN was found to be 99.1 % accuracy	53
Linear Regression, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Multi-Layer Perceptron, and Naïve Bayes.		FibrAlm has proven to be highly effective for diagnosing patients at risk of liver early steatosis and fibrosis	54
Hybrid Feature Selection and Deep Learning Framework combining Random Forests with Principal Component Analysis (PCA) and L1 regularization	Thyroid	Random Forest classifier obtained the maximum accuracy of 96.30 %	55
Logistic regression, Random Forest, and k-nearest neighbors (KNN)	Diabetes	KNN model obtained the maximum accuracy of 96.09 %, a sensitivity of 98.54 %, and a specificity of 93.63 %	56
Ensemble approach named En-RfRsK integrating Random Forest (RF), radial support vector machine (R-SVM), and K-nearest neighbor (KNN).		The accuracy of the En-RfRsK approach is 88.89 %	57
Support vector machine, logistic regression (LR), K-nearest neighbor, and random forest.	Hepatitis	LR model, when integrated with the SMOTE technique, achieved the highest accuracy level of 93.18 %	58
Random Forest, Cat Boost, Bagging Classifier, SGD Classifier, Gaussian NB, Bernoulli NB, Multinomial NB, Linear Discriminant Analysis, ANN, and MLP.		Random Forest classifier achieved an accuracy of 99.53 %	59
Decision Tree, Random Forest, Support Vector Machine, Gradient Boosting, and Voting classifiers	Alzheimer's Disease	Random forest achieved the highest accuracy of 86.92 %	60
Deep learning algorithm like ResNet-50v2		ResNet-50v2 gives the best accuracy of 91.84 % and f1-score of 0.97 for AD class.	61
Hybrid Deep Learning based on convolutional neural network and XGBoost	Lungs Cancer	The proposed method achieved an accuracy of 97.43 %	62
Random Forest		The results demonstrate a model accuracy of roughly 97.9 %.	63
Deep learning based modified ResNet50-based architecture integrating faster R-CNN and dual optimizers	Prostate Cancer	The proposed model obtains high accuracy of 95.24 %	64
A novel deep learning-based 3D convolutional neural network		The results demonstrated an accuracy of 87 %.	65
Deep CNN and fine-tuned VGG16 and Xception models	Brain tumor	Deep CNN attained an accuracy of 97.27 %	66
Ensemble Deep Neural Support Vector Machine (EDN-SVM) classifier		The model obtained an accuracy of 97.93 %,	67
K Nearest Neighbor (KNN), decision trees (DT), Convolutional Neural Network (CNN), Visual Geometry Group (VGG-16, VGG-19), ResNet50, DenseNet201, Xception, and InceptionV3	Covid 19	VGG-19 performed best with a training accuracy of 99 %, testing accuracy of 98 %	68
Decision trees (DT), K-nearest neighbors (KNN), support vector machines (SVM), and Naïve Bayes		The suggested model achieved maximum 99.9 % accuracy using SVM for severe COVID-19 infection	69

mild, moderate, and severe. Four machine learning techniques—decision trees (DT), K-nearest neighbors (KNN), support vector machines (SVM), and Naïve Bayes—are employed for the classification of scans. The proposed model attains accuracy rates of 99.12 %, 98.24 %, 98.73 %, and 99.9 % for COVID-19 infection segmentation across the normal, mild, moderate, and severe stages, respectively.⁶⁹ Table 1 provides the overview of recent studies using ML for diagnosis of different diseases.

In Table 2, Machine learning applications are given where algorithms are utilized to predict the onset of diseases.

Supervised learning has proven to be highly effective in the diagnosis and prediction of various diseases, including cancer, cardiovascular disorders, neurodegenerative conditions, and infectious diseases. Based on the literature survey, ensemble models, and deep learning architectures have significantly improved diagnostic and predictive performance. Although supervised learning demonstrates high accuracy in well-labeled datasets, its reliance on extensive, high-quality annotated data constrains scalability in the context of rare diseases or underrepresented populations. Utilizing hybrid, semi-supervised, or transfer learning methodologies presents a significant challenge for future research endeavors.

3.1.2. Unsupervised learning in segmentation, clustering and anomaly detection in medical imaging and patient data analysis

Recent advances in machine learning have enabled automated medical image analysis systems to detect tumors or lesions caused by neuro-degenerative disorders with remarkable efficiency. Unsupervised learning, a type of machine learning algorithm involves training a system with unclassified and unlabeled information, enabling the algorithm to make decisions without any explicit instructions.^{78–80} Unsupervised learning algorithm is employed to analyze MRI images in the absence of labeled training data. Brain tumor segmentation in magnetic resonance (MR) images facilitates quantitative assessment of brain tumors and offers essential insights for improved comprehension of tumor characteristics and treatment planning. Brain tumor segmentation can be achieved through traditional image processing methods. Nonetheless, the precision of these methods are constrained. Deep learning methods have been utilized for brain tumor segmentation, achieving remarkable accuracy in segmentation outcomes.⁸¹ Unsupervised learning methods have been investigated for brain tumor segmentation, eliminating the need for manual annotation. A significant number of these unsupervised approaches utilize reconstruction-based methods.⁸² Synthetic data generation has recently been investigated for unsupervised tumor segmentation⁸³. Nevertheless, the generation of unrealistic tumors for training segmentation models may result in suboptimal performance, attributable to the disparity between actual and synthetic brain tumor MR images. Therefore, the creation of synthetic tumors for unsupervised brain research. Some researchers have proposed a two-stage synthesis strategy named SynthTumour for unsupervised segmentation of brain tumors in MR images. Their study examined methods for synthesizing more realistic brain tumor data and strategies to prevent model overfitting⁸⁴. Initially, a junior segmentation model is developed, wherein researchers create shape and intensity models to produce synthetic brain tumors for the purpose of model training. In the subsequent stage, an advanced model is developed to improve segmentation performance. The proposed method has been validated using both public and in-house datasets for brain tumor segmentation. Fig. 3 illustrates an Unsupervised Brain tumor Segmentation Technique with two stage synthesis.

Furthermore, unsupervised learning methods are extensively employed for clustering biomedical images. The cost associated with manual annotations of medical data, in terms of time and resources, is significantly high and serves as a bottleneck in the implementation of supervised learning algorithms. Thus, unsupervised learning methods may serve as automatic annotation tools, with the resulting clusters utilized as labels for training classifiers in a supervised manner.⁸⁵ Some researchers proposed an efficient unsupervised learning pipeline utilizing compressed pre-trained deep features. The method for unsupervised learning of biological images utilizes ensembles of features derived from pre-trained transformers and CNNs, which are subsequently compressed using a customized

Table 2
Machine learning applications for predicting the onset of diseases.

Algorithms	Disease Prediction	Finding	Ref.	Year
Logistic regression (LR), Decision tree (DT), Random Forest (RF), and Extreme Gradient Boosting (XGB)	Onset of diabetes	RF was the most pertinent technique for hypertension, myocardial issues, and retinopathy	70	2024
ANN, k-NN, SVM, DT, RF, XGboost, and Adaboost	Breast Cancer	XGboost model obtained the maximum recall of over 96 %	71	2024
Hybrid deep learning based MDensNet201-IDRSRNet	Heart disease	The proposed model achieved an accuracy of 99.12 %	72	2024
Adaptive data preprocessing and ensemble model	Liver Disease	The model showed significant training and testing accuracy, achieving 99.87 % and 99.80 %, respectively.	73	2024
8 Ensemble methods	Kidney Disease	LightGBM demonstrated better performance compared to state-of-the-art and other ensemble methods	74	2024
Gradient-Boosted Trees (GBT), Light Gradient-Boosting Machine (LightGBM), Random Forest (RF), eXtreme Gradient-Boosting (XGBoost), Logistic Regression (LR), Adaptive Boosting (AdaBoost)	Alzheimer's Disease	GBT model achieved the best AUC-ROC scores of 0.809–0.833	75	2023
A hybrid method incorporating feature selection via the Antlion Optimization Algorithm (ALO) and classification through Random Forest (RF) along with the XG-Boost	Dengue disease	The proposed framework attained a high level of precision of 96.17 %, a precision of 93.95 %	76	2024
Ensemble of the Random Forest and XGboost models	Lung Cancer	The proposed model achieved an accuracy of 71.2 %	77	2024

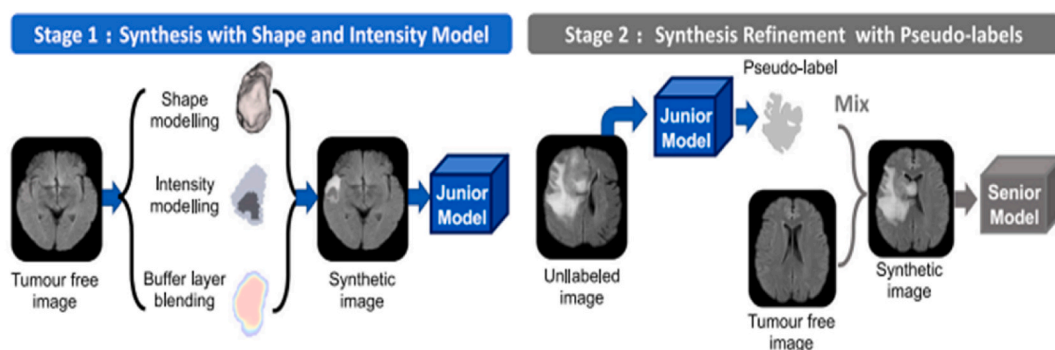


Fig. 3. An Unsupervised Brain tumor Segmentation Technique with two stage synthesis⁸⁴.

AutoEncoder to identify significant patterns from deep pre-trained features. The proposed method demonstrates superior clustering accuracy compared to existing state-of-the-art results for biological datasets across various domains.⁸⁶

Anomaly detection in medical imaging has proven to be crucial in medical image analysis. The Unsupervised Anomaly Detection (UAD) method aims to differentiate anomalous tissue from normal anatomy by learning from healthy structures. Generative models are frequently employed to generate counterfactual healthy samples from diseased images, facilitating anomaly detection without the need for pixel-level annotations.⁸⁷ The use of such models removes the necessity for extensive manual labeling, which can present difficulties. Additionally, it enables the detection of unseen or rare anomaly cases that are underrepresented in the dataset for fully supervised models. However, the inability of these models to reconstruct unhealthy structures results in reconstruction errors that signify anomalies. Nonetheless, a considerable challenge lies in achieving a balance between the precise reconstruction of healthy anatomy and the inadvertent replication of abnormal structures. In recent times, diffusion models have been proven to demonstrate potential for detailed and accurate reconstructions. Some researchers suggested conditioning the denoising process of diffusion models using supplementary information obtained from a latent representation of the input image.⁸⁸ Other researchers have proposed an unsupervised anomaly detection model specifically for medical images. The Synomaly noise function was applied to improve the efficacy of anomaly detection. This noise is introduced to healthy training images to replicate the characteristics of actual anomalous tissues following the noising process. A multi-stage diffusion process was proposed for the inference phase. Extensive validation was conducted across various medical imaging modalities, including brain MRI, liver CT, and carotid ultrasound (US) datasets. The findings from the three predominant medical imaging modalities (ultrasound, computed tomography, and magnetic resonance imaging) illustrate the applicability of the proposed method across various contexts. Although unsupervised anomaly detection offers benefits, its effectiveness is significantly influenced by the generative model's capacity to maintain healthy structures and precisely reconstruct pathological regions as their healthy equivalents. This highlights the necessity for additional investigation into advanced generative models to improve performance.⁸⁹

Unsupervised learning has become a critical tool for medical imaging, particularly in tumor segmentation, anomaly detection, and biomedical image clustering, as it mitigates the dependency on costly and time-consuming manual annotations. By minimizing the domain gap between synthetic and real data can enhance unsupervised segmentation performance. Furthermore, leveraging diffusion models or hybrid generative frameworks can ensure more accurate reconstruction of healthy anatomy and precise anomaly detection. Further development of explainable metrics can be carried out to quantify how well models separate normal and pathological tissue. Also, more focus on realistic data synthesis, multi-modal learning, and clinically interpretable generative models to maximize impact. Also, training the model to selectively identify specific types of anomalies while ignoring others can facilitate selective segmentation. These advancements would enable the model to detect and classify anomalies, thereby enhancing its value in clinical decision-making.

3.1.3. Reinforcement learning in personalized treatment plans & robotics-assisted surgery

Providing personalized recommendations has become crucial in the medical field for lessening the impact of overloaded information. The biomedical sector has yet to fully explore proper recommendations. The health care industry typically addresses chronic or enduring concerns. When it comes to tackling health behaviors and chronic diseases, the capacity of individuals to manage their programs is of the utmost importance. Participating in physical activities, adhering to prescribed medications, and following eating plans are all examples of common self-management behaviors. To assist in such endeavors, it is essential to take into account the distinct lifestyles and priorities of individuals, as well as their psychological and psychosocial circumstances, and environmental influences.⁹⁰ Conventional recommended systems do not take into account the long-term consequences, thereby failing to capture the changing emotions of individuals. When compared to non-individualized treatments, some studies have shown that providing personalized self-management support for common health conditions or chronic diseases yields better results.^{91,92} The majority of current recommended systems have not fully addressed the problems of data sparsity and non-linear user-item connections. Reinforcement learning systems have the potential to alter recommendation techniques.⁹³ Reinforcement learning is an approach for guiding agent behavior within an environment to optimize cumulative rewards. Recent studies have focused on various disease treatment recommendation systems utilizing reinforcement learning. Some researchers sought to create a reinforcement learning-based treatment recommendation model utilizing the health records of South Korean citizens to support physicians. This

study is significant as it demonstrates how expert and intelligent systems effectively address clinical challenges in prescribing appropriate diabetes medication based on the physical condition of patients with diabetes.⁹⁴ Others introduced an intelligent multi-agent system utilizing algorithms from both Reinforcement Learning and Deep Learning to offer automated support to patients adhering to a prescribed home treatment regimen. The system provides patients with personalized reminders for medication intake and alerts to prevent potential medication errors. The findings indicate that the system enhances the quality of home assistance services by decreasing medication errors.⁹⁵ Another study suggested a robust and effective model-free method based on deep reinforcement learning (DRL) for the closed-loop control of drug dosing in cancer chemotherapy. A nonlinear pharmacological model for cancer is employed to simulate patient responses and capture the dynamics of cancer progression. The findings indicate the superiority of the approach regarding the process and duration of cancer chemotherapy treatment.⁹⁶

Recent research indicates that deep reinforcement learning (DRL) enhances medication dosing, adherence reminders, treatment planning and home care support, surpassing conventional recommendation or rule-based systems in accuracy, patient safety, and treatment efficiency. The integration of multi-modal data can be attained by combining electronic health records, wearable sensors, and patient-reported outcomes to enhance context-aware, personalized recommendations. Further research may elucidate the design of DRL frameworks capable of coordinating various treatment aspects concurrently (e.g., medications, lifestyle interventions, therapy schedules).

Furthermore, reinforcement learning techniques have demonstrated effective outcomes in automating sub-tasks within robotic surgery systems.^{97,98} Recent advancements in Robot-Assisted Minimally Invasive Surgery (RAMIS) have greatly benefited from the incorporation of reinforcement learning (RL) techniques.⁹⁹ These advancements have improved surgical accuracy, safety, and independence. Surgical robots have demonstrated effective performance, enabling their application in complex and high-risk environments, thus it alleviates stress on the surgeon and enhances surgical accuracy. Researchers devised a deep reinforcement learning approach to regulate the gripper arm during the cutting of soft tissue within a continuous action space. The proposed method demonstrated superior performance compared to the leading technique in the soft pattern cutting task regarding accuracy.¹⁰⁰ Furthermore, The SaferPlan system utilized deep reinforcement learning to enhance screw trajectory planning in robotic surgeries. A recent study introduces SaferPlan, a deep reinforcement learning-based system aimed at optimizing screw trajectory planning in robotic surgeries. The method focuses on safety via distance-based filters.¹⁰¹ Furthermore, Multi-Agent Reinforcement Learning (MARL) is increasingly recognized as an effective method in robotic surgery, especially for the simultaneous coordination of multiple robotic instruments or subsystems. MARL facilitates the learning of optimal policies by distributed agents via collaboration or competition, thereby improving task efficiency, accuracy, and adaptability in intricate surgical settings.¹⁰²

Based on the review it is apparent that reinforcement learning (RL), encompassing deep RL and Multi-Agent RL (MARL), has markedly improved Robot-Assisted Minimally Invasive Surgery (RAMIS) through the automation of sub-tasks, enhancement of accuracy, and alleviation of surgeon stress. It is important to know that developing methods that incorporate safety limits to mitigate intraoperative errors is crucial. The integration of surgical expertise with reinforcement learning agents represents a potential avenue for semi-autonomous surgical procedures.

3.1.4. Deep learning in image recognition, natural language processing & predictive analytics

The unique challenges of medical records when compared to general domain text include the following: the heavy use of acronyms and unusual clinical terms by healthcare providers; inconsistent document organization and structure etc. The rapid growth and availability of unorganized data since the inception of the Internet have had a significant role in the establishment of natural language processing (NLP) as an essential technology for tackling complex and knowledge-based tasks. Some of the steps involved in Natural Language Processing (NLP) include morphological analysis, syntactic analysis, semantic analysis, discourse processing etc..^{103,104}

Conventional machine learning approaches have been extensively employed in medical natural language processing tasks. While these methods necessitate considerable manual effort and continue to exhibit limitations in accuracy, deep learning (DL) based methods have demonstrated substantial advancements.¹⁰⁵ Deep learning is a subfield of machine learning that builds hierarchical representations of input using multi-layer neural network designs.¹⁰⁶ Deep learning is mostly used in the healthcare sector to apply advanced computer vision methods as well as image processing to medical scans to diagnose disorders. Image analysis and Electronic Medical Records (EMR) natural language processing are two clinical applications where deep learning has shown significant promise. The utilization of Natural Language Processing (NLP) in analyzing free-text inside Electronic Medical Records (EMR) holds great potential due to the comprehensive nature of EMRs. These records encompass clinical assessments, overall background, diagnostic test summaries and research, as well as longitudinal monitoring of the patient's progress. Several deep learning architectures have employed adult electronic medical records (EMRs) to extract features and perform natural language processing (NLP)^{107–111}.

Some researchers have applied deep learning-based natural language processing methods to detect colonoscopic information. The study compared long short-term memory (LSTM) and BioBERT models to identify the one exhibiting superior performance for colonoscopy reports, ultimately determining that the bidirectional LSTM with conditional random fields was the most effective. The NLP model utilizing pre-trained word embeddings outperformed the one employing one-hot encoding across the majority of labels. The method employed in this study yielded promising results, indicating its applicability to diverse practical applications.¹¹² Others sought to predict the progression of glaucoma necessitating surgery through deep learning techniques applied to electronic health records. This analysis incorporated features derived from both structured clinical data and natural language processing of unstructured clinical text.¹¹³ Another study was conducted to design and validate large language models (LLMs) for the purpose of automatic clinical

diagnoses. Large language models represent the most effective methodology in natural language processing. Classification models were developed by fine-tuning BERT, a pre-trained transformer-based deep learning model. The models were developed to recognize 12 medical symptoms and 2 patient histories from simulated clinician–patient interactions across 6 primary symptom scenarios in emergency triage settings. The models were subsequently analyzed utilizing eXplainable artificial intelligence (XAI) and the Shapley additive explanation (SHAP) method. XAI utilizing SHAP offered dependable explanations for model outputs.¹¹⁴

Predictive analytics is an essential requirement in the healthcare sector in recent times. Accurate and timely predictions can potentially save lives, while inaccurate predictions can pose a risk to patients' lives. Therefore, it is crucial to have dependable and effective techniques for healthcare predictive analysis. However, the increasing complexity and volume of healthcare data pose significant challenges for traditional machine learning methods, particularly in handling high-dimensional features and heterogeneous data sources. Deep learning has emerged as a superior alternative, delivering robust predictive performance. The expansion of big data has spurred the development of rapid and precise analytics methods.^{115,116}

Some researchers examined the efficacy of deep learning, particularly convolutional neural networks (CNNs), in conducting the predictive analytics of heart disease risk through critical personal health indicators. The model employed embedding layers to convert categorical data into numerical forms, convolutional layers to identify spatial features, and dense layers to capture complex interactions and predict cardiovascular disease risk. The model demonstrates significant potential as a valuable resource for clinicians in the prevention and management of cardiovascular disease.¹¹⁷ Others employed a predictive analytics methodology for stroke prediction, assessing multiple factors within electronic health records. Through the application of diverse statistical techniques and principal component analysis, the study identified age, heart disease, average glucose level, and hypertension as the most significant factors in stroke prediction. A perceptron neural network utilizing these four attributes achieves the highest accuracy rate and the lowest miss rate when compared to the use of all available input features and other benchmarking algorithms.¹¹⁸

Deep learning has significantly progressed healthcare applications like image recognition, natural language processing of electronic medical records (EMR), and predictive analytics. Most importantly, the incorporation of explainable AI (XAI) methods like SHAP and LIME have proven to improve the interpretability of these models significantly. Future research should concentrate on domain-adaptive large language models (LLMs) for electronic medical record (EMR) comprehension, which can learn from multimodal healthcare data, encompassing text, imaging, and genomics. The importance of interpretable and explainable predictive models is essential for clinical adoption.

3.2. IoT in healthcare

With the growth of the health care industry, there is a demand for convenient data transfer and efficient management of data which is facilitated by IoT. The IoT connected medical technologies allow physicians to remotely monitor patients' health status from any location.

3.2.1. IoT-based wearable devices

In this fast-changing world, it has become imperative for the majority of technologies to be interconnected, accessible from a distance, and capable of being analyzed. By linking physical objects to the Internet, the Internet of Things (IoT) enables the development of Internet-enabled intelligence in consumer electronics. The Internet of Things (IoT) is connected using computers equipped with sensors, microcontrollers, and transceivers.^{119,120} The implementation of the Internet of Things (IoT) in healthcare holds great potential for enhancing existing technologies and decreasing costs across several healthcare domains.¹²¹ Wearable sensor devices enabled by the Internet of Things are facilitating tremendous development and expansion in the healthcare industry.

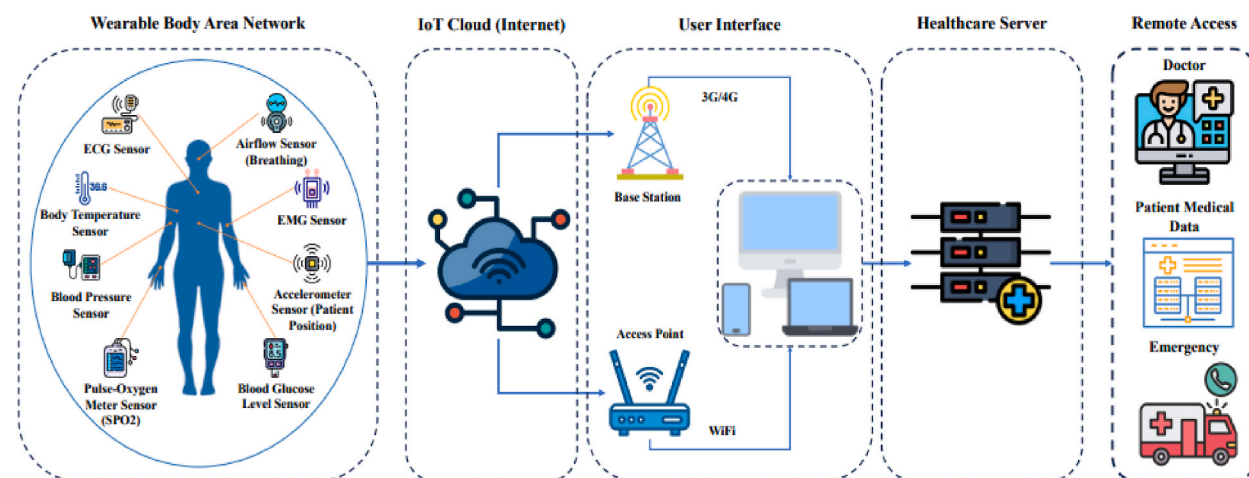


Fig. 4. Five-layer communication of wireless body area network through IoT cloud.

To guarantee a rapid and smooth transmission of data, various communication protocols exist. The data can also be utilized for analysis through the application of data analytics. A Wireless Sensor Network (WSN) is a network consisting of sensors that are linked and communicate wirelessly using certain protocols. Researchers conducted a concise comparison of several deep learning methods and the use of Wireless Sensor Networks (WSN) to assess and enhance the Internet of Things (IoT) in the healthcare sector.¹²²

In order to provide ongoing patient monitoring and intelligent care delivery, smart hospital ecosystems are increasingly based on wearable IoT devices, seamless integration with electronic health records (EHRs), and real-time cloud connectivity. An AI-enabled wearable IoMT architecture for long-term care settings was presented by the researchers. It combined intelligent upstream processing with edge-based physiological signal collecting. Vital signs and behavioral data are sent by wearable sensors to edge nodes, which conduct instantaneous analytics before sending processed insights and organized EHR data via secure cloud pipelines. By utilizing wearable data and connected, high-fidelity digital patient files, this architecture enables tailored care coordination, minimizes needless hospital stays, and enables real-time clinician warnings.¹²³

3.2.2. Integration of smart IoT sensors in medical devices

Every health care system that utilizes Internet of Things (IoT) technology incorporates a layer of sensors. These sensors gather data from users/patients by detecting their vital signs and other essential signals. The data is subsequently transformed into a form that can be analyzed and tracked. This layer may consist of a variety of sensors capable of utilizing sensors to identify and track physical movement and observe essential health indicators, and other distinct signals that may suggest an irregularity. Multiple prototypes have been constructed utilizing various types of sensors.¹²⁴

These sensors are primarily intended to be worn, as users must be able to perform their usual activities with no disturbance. Several of these systems employ Body Area Networks (BAN) or Wireless Sensor Networks (WSN) for networking through IoT cloud. Fig. 4 shows the commonly used architecture of remote monitoring and WBAN communication through IoT cloud.

As illustrated in Fig. 4, the WBAN is capable of diagnosing many human ailments and monitoring various user attributes, such as body temperature, heart rate, and blood pressure. The smart phone can then get the data from the wireless sensors once the anomalous state has been determined. After that, this information is sent to doctors or any other assistance desk so they can take the appropriate action. As seen in Fig. 4, the data can be delivered from the gateway to the destination over the Internet or a cellular network. WBANs are also utilized for remote monitoring and preventative health actions. Furthermore, the field also encompasses microcontrollers such as Arduino, STM32 Microcontroller, Intel Galileo, Raspberry Pi etc..^{125–127} The communication protocols utilized were MQTT, LoRaWAN, GSM, ZigBee, BLE, and GPRS.^{128–130} In order to facilitate disease prediction and risk stratification, wearable and bedside sensors transmit data to IoT clouds for preprocessing, feature extraction, and model training. Home monitoring data is streamed, cleaned, processed, classifiers are trained, and doctors are given predictions for post-discharge risk monitoring by one cloud-based machine learning platform.¹³¹ IoT/IoMT-ML pipelines enhance cardiovascular care predictability and continuous monitoring, including event prediction, heart failure, and arrhythmia.¹³² The smart IoT sensors include respiratory sensor,^{133,134} heart Beat Monitoring Sensor,^{135–139} activity detection sensors,^{140–144} accelerometer sensor,¹⁴⁵ blood pressure sensor,^{146,147} blood glucose monitoring sensor,¹⁴⁸ body temperature sensor,^{149,150} EMG sensor^{151,152} etc.

3.2.3. Remote monitoring systems

Remote patient monitoring, also known as telemonitoring, focuses on specific sub-groups of patients, including those with chronic illnesses, mobility challenges, disabilities, post-surgery patients, neonates, and elderly patients. It is advisable to continuously monitor all patients with these types of illnesses. Most research adheres to the principle of offering patients the ability to move and engage in activities in their own homes or personal environments, which is more advantageous than being confined to an expensive hospital room. People in their elder age no longer require constant supervision from a caregiver because of cutting-edge remote health monitoring software. Patients who have suffered accidents or sudden injuries may only be monitored remotely while they are being taken to the hospital in an ambulance. Nevertheless, the main emphasis has been on guaranteeing a secure journey to the hospital, while remote monitoring assists in delivering timely medical interventions for highly critical patients. The doctors can see and track the decline or preservation of the patient's condition, while also providing guidance to the paramedics who stay with the patients, if needed.¹⁵³

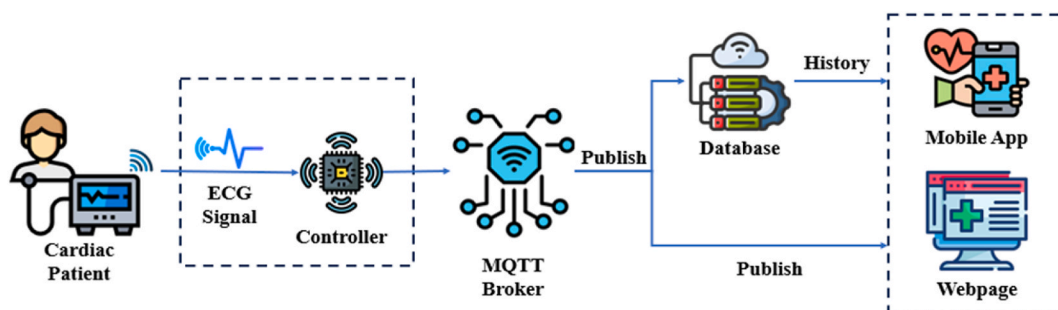


Fig. 5. Architecture of remote patient monitoring system.

Sensor-based data collection, processing units (usually edge or cloud-based), hospital-side interface terminals, and the communication network that links these components make up the four main parts of a contemporary remote patient monitoring (RPM) structure.¹⁵⁴ A data acquisition system is made up of a number of sensors or devices that are equipped with embedded sensors that are capable of transmitting data for wireless transmission. Following the transmission of the gathered data and conclusions to a health care professional who is connected to the system through the communication network, the underlying communication network establishes a connection between the data acquisition system and the data processing system. They may be instructed to take specific first-aid measures and/or prescribed medications. Remote patient monitoring devices are specifically intended to collect various physiological data from patients. The most prevalent types of data monitored remotely include Electrocardiogram (ECG), Electroencephalogram (EEG), cardiac rhythm and respiratory rate, blood oxygen saturation or pulse oximetry, blood pressure, neural signals, body/skin temperature, and blood glucose level. Furthermore, the weight, activity level, and sleep data of the patient are occasionally gathered. The utilization of IoT enabled remote monitoring systems has significantly advanced the management of different diseases. In order to ensure the reliability of the real-time electrocardiogram (ECG), researchers built a remote patient monitoring system that is based on the Internet of Things (IoT).¹⁵⁵ As shown in Fig. 5, the four modules that make up the proposed patient monitoring system are a webserver, a communication module, a main controller, and a sensor. Electrocardiogram (ECG) data is received from the patient's heart by means of the AD8232 ECG sensor. The WiFi-enabled Arduino ESP 32 serves as the system's primary controller in this work. The main controller processes the collected ECG data in real time and then uses the WiFi module to communicate with the MQTT broker. The clinician can view the ECG data in real-time or from past sessions by accessing the web server from a computer or smartphone.¹⁵⁶ Fig. 5 shows remote patient monitoring that improves patient-physician communication, makes early illness decompensation and intervention possible, and aids patient education.

Recently, some researchers developed a compact, non-invasive IoT-based healthcare kit for the continuous monitoring of vital signs, including SpO₂, temperature, heart rate, and respiration rate. The system provides a portable solution for personal health monitoring, allowing remote access to health data and promoting early detection of potential health issues.¹⁵⁷

3.2.4. Inter-operability for seamless data transfer

The potential advantages of cutting-edge disciplines like machine learning and big data analytics are limited by this underutilization. Although 36 % of American individuals used wearable technology, according to a recent cross-sectional survey, only roughly 25 % shared their data with healthcare professionals, even though over 78 % said they would be ready to do so. This suggests that there are gaps in workflow alignment and integration.¹⁵⁸ Critical obstacles to using patient-generated health data (PGHD) for preventive healthcare were also highlighted by another study that interviewed clinicians managing ischemic heart disease and discovered that interoperability, clinical workload, and data reliability issues hindered effective use of PGHD.¹⁵⁹

A significant barrier to interoperability, according to researchers, is the lack of commonly accepted, uniform standards for gathering and sharing clinical and patient-generated data; even in cases where frameworks such as FHIR are in place, heterogeneous vocabularies, inconsistent metadata, and compartmentalized system architectures continue to obstruct smooth data sharing.¹⁶⁰ Additionally, a large portion of wearable and remote monitoring data is underutilized in decision-making due to limited integration into clinician workflows and EHRs.¹⁶¹ Similar to the layered approach in the OSI model, the architecture of contemporary IoT healthcare implementations frequently extends from physical sensors to edge gateways and cloud services. But the computational, memory, and energy limitations of resource-constrained devices, particularly sensor nodes, make it difficult for them to preprocess, store, or perform important logic on-site.¹⁶² According to a study on smart healthcare networks, raw sensing is still lightweight and only generates data, and such limited devices must rely on edge or fog intermediaries to carry out real-time analytics, security tasks, and communication protocol translation.¹⁶³ This layered design reinforces the hybrid hierarchy common in IoT systems by highlighting the need to transfer complex activities to more capable infrastructure.

Fast Healthcare Interoperability Resources (FHIR) Standard is an extensively used standard in healthcare. Standardized language, standards, and protocols independent of vendor platforms are advocated for by the International Healthcare Technology Standards Developing Organization (IHTSDO). As an example, one format used for sharing information across versions is Health Level Seven (HL7). Clinical Document Architecture (CDA) and other prior HL7 standards were integrated to form FHIR. HTTP, RESTful, XML, and JSON are some of the most popular web protocols that FHIR incorporates these features with.^{164,165} Moreover, Ambient Assisted Living (UniversAAL) is another standard. The well-being and autonomy of older individuals have been greatly enhanced by recent developments in Ambient Assisted Living (AAL).^{166–168}

The incorporation of IoT and wearable devices in healthcare facilitates continuous remote patient monitoring, real-time data collection, and personalized care via WBANs and IoT-ML pipelines that enhance predictive analytics and risk stratification. Remote monitoring systems improve patient autonomy, decrease hospital visits, and enable prompt interventions. However, some challenges exist, including limited interoperability among diverse devices and electronic health record systems, computational limitations on resource-constrained IoT devices, concerns regarding data privacy and security, and inadequate integration of wearable data into clinical workflows. Researchers need to focus on the development of standardized frameworks for the integration of devices and electronic health records, the implementation of edge-AI for real-time analytics, the adoption of privacy-preserving machine learning methods like federated learning, the design of patient-centered devices focused on usability and feedback, and the utilization of AI-driven insights to facilitate predictive and preventive healthcare interventions.

3.3. Case studies on integrated ML-IoT in healthcare

This section provides an overview of the recent research works focusing on employing different diagnosis use cases that have been proposed or implemented using machine learning (ML) and the Internet of Things (IoT) platforms. A diagnosis is established by examining the patient's symptoms, which are monitored by sensors to identify the kind of disease they are currently dealing with.

3.3.1. Case studies on predictive analytics for diabetes management

Doctors can leverage IoT-enabled cloud systems with AI and ML to monitor patients' physical activity and analyze data from wearable sensors for continuous glucose monitoring (CGM), and lifestyle inputs. These systems integrate real-time glucose levels, activity, meals, insulin doses, and sleep patterns to predict blood glucose trends and support timely clinical interventions. For type 1 diabetes, CGM systems combined with edge-based ML allow near-real-time glycemic predictions up to 45 min in advance, aiding insulin adjustments and hypoglycemia prevention without relying solely on the cloud.¹⁶⁹ Meta-analyses show Random Forest offers the most accurate forecasts over 15–60 min, with rising RMSE values but still within therapeutic use.¹⁷⁰ Frameworks like SmartEdge further improve latency and prediction accuracy by about 5 % compared to cloud-only solutions.¹⁷¹ Hybrid systems increasingly incorporate blockchain to ensure secure, inter-institutional data sharing. AI-driven models enhance diabetes risk prediction by 4–6 % compared to baseline approaches, highlighting the growing role of AI, ML, and IoT in delivering personalized, predictive diabetes care.^{172,173}

For longer prediction horizons and up to 10 % higher accuracy than LSTM baselines are supported by models such as AttenGluco, which use transformer architectures to fuse wearable sensor data with CGM measurements.^{174,175} Similarly, with RMSE as low as 5.9–11.3 mg/dL, large-scale ANN models trained on millions of minutes of T1D data perform better than traditional meters. However, they struggle with rapid glucose changes because of CGM latency.^{176,177} Additionally, IoT integration makes non-invasive monitoring possible through NIRS devices, which gather glucose data every half hour, transmit it to the cloud, set off alarms for hypoglycemia (less than 4 mmol/L) or hyperglycemia (more than 11 mmol/L), and enable remote monitoring via SMS alerts, enhancing safety, particularly at night.^{178–182}

Meanwhile, other body fluids including saliva, sweat, tears, or skin are being used to develop non-invasive glucose monitoring technologies like Raman spectroscopy, NIRS, mid-infrared, fluorescence, and bioimpedance.¹⁸³ Prediction accuracy has increased through developments that combine wearable optical and electrochemical sensors with AI and ML models (CNNs, Random Forest, XGBoost). A study found that XGBoost with multimodal physiological sensors decreased errors (RMSE ~11.9 mg/dL, MARD ~7.1 %),¹⁸⁴ while CNN and Random Forest with SWIR spectroscopy produced great precision (MAPE 4.82 %, 100 % Clarke Zone-A).¹⁸⁵ Predictive features for upcoming hyperglycemia events are still absent from the majority of systems, though.¹⁸⁶

3.3.2. Case studies on remote patient monitoring for heart disease

Since cardiovascular disease is still one of the top causes of death, wearable sensors, cloud/fog computing, and artificial intelligence (AI) are being used in IoT-enabled remote patient monitoring (RPM) systems to provide real-time cardiac care. While ML classifiers (Random Forest, CNN, k-NN) with ECG sensors demonstrated 94 % accuracy for premature ventricular contractions and 97 % accuracy for normal beats, advanced models like ensemble CNN–LSTM have achieved 97.23 % accuracy in detecting abnormalities in multiple vital signs of heart failure patients, guaranteeing prompt alerts to doctors.^{187,188} These systems enable low-latency, high-fidelity monitoring, even in resource-limited settings, reducing rehospitalization rates and supporting predictive cardiac care. Wireless Sensor

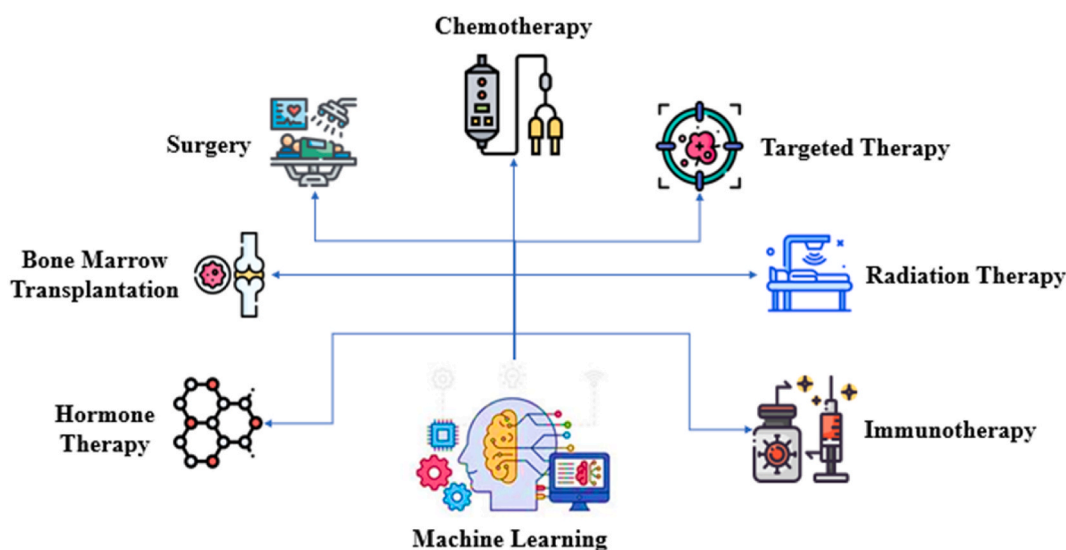


Fig. 6. Cancer treatment planning.

Networks (WSN) further enhance monitoring by continuously tracking temperature, heart rate, pulse, blood pressure, and ECG signals for critical patients. ECG data, essential for diagnosing arrhythmia, hypertension effects, or heart attacks, is transmitted to the cloud for physician access, while caretakers receive alerts in case of abnormalities.^{189,190} Following procedures like data preprocessing, classifier evaluation (SVM, KNN, NB, DT, RFC), and hardware prototyping with cloud connectivity, Divya et al. (2024)¹⁹¹ and others^{192–196} created IoT-based wearable healthcare systems that incorporate machine learning to continuously monitor vital signs like temperature, pulse, and ECG for real-time cardiac disease detection. By facilitating real-time data analysis, early diagnosis, remote monitoring, and disease prevention, these IoT-driven solutions are revolutionizing cardiac care and lowering the growing burden of heart illnesses and related medical expenses.¹⁹⁷ In order to help doctors provide prompt interventions, adjust treatments, and enhance care coordination, these systems gather and analyze patient data from clinics, hospitals, and homes using sophisticated sensors, communication networks, and artificial intelligence algorithms.^{198–200} The efficacy of IoT-AI integration in real-time prediction, complication prevention, and cost-effective cardiovascular management across clinical and residential settings was demonstrated in a recent study. Further, it validated this approach by combining wearable blood pressure sensors, cloud analytics, and AI classifiers (Naïve Bayes, KNN, Random Forest, and SVM). This study achieved 99.44 % accuracy in early hypertension detection.²⁰¹

3.3.3. Case studies on personalized cancer treatment

When it comes to the number of fatalities, the most common type is lung cancer. The process of diagnosing cancer can be greatly accelerated by using computer-aided diagnostic (CAD).²⁰² Because of this, machine learning is becoming more and more important in the detection and treatment of different forms of cancer. AI in cancer is expected to speed up the procedures of diagnosis and treatment planning.²⁰³ Oncology uses ML for medication synthesis at the point of care, treatment planning, medical image analysis, and patient survival prediction.²⁰⁴ Prognostic prediction for patients has already been enhanced by the ML model over conventional staging techniques. The implementation of ML could also have a significant positive impact on the synthesis done directly of the required medicinal compounds when receiving treatment.

Planning a course of treatment for cancer is extremely complicated and requires interdisciplinary cooperation from radiologists, surgeons, oncologists, data scientists, and rehabilitation experts. Personalized oncology care is being revolutionized by recent developments in AI, DL, and ML, which are frequently combined with IoT-connected high-resolution imaging, sensor devices, and omics platforms. The types of cancer treatments that facilitate treatment planning and help prevent cancer recurrences are depicted in Fig. 6.

While ML-IoT systems with wearable sensors and ensemble models (Random Forest, XGBoost) support real-time monitoring, adaptive treatment adjustments, and early relapse detection, methods such as DRL have demonstrated strong potential in automated radiation planning, including prostate cancer IMRT and HDR brachytherapy.^{205–207} In order to improve tailored alternatives like immunotherapies and targeted medications, advanced ML/DL models (CNNs, autoencoders, transformers) use clinical, imaging, and multi-omics data to predict therapy response and survival with >90 % accuracy.²⁰⁸ Gleason grading, lesion segmentation, recurrence prediction, tumor classification, and the identification of treatments with fewer adverse effects are all further improved by AI-powered MRI, pathology, and genomic methods.^{209,210} Notwithstanding these developments, computational modeling still faces difficulties since ML needs to address deeper biological causality than supervised imaging classification and single-source omics.^{211,212} However, by automating repetitive processes and improving clinical decision-making, ML-IoT healthcare systems trained on huge datasets can promote precision oncology, decrease human error, improve efficiency, and assist risk assessment and treatment planning.^{213–217}

Based on the recent studies, together with IoT data, machine learning, deep learning models, ensemble models, multi-omics analysis, and AI-enhanced imaging can provide precise, tailored, and adaptive medical interventions. Thus, patient outcomes are improved, and complications and readmission rates are lowered as well. To guarantee accurate, useful insights, non-invasive monitoring methods and multi-modal IoT data must be standardized and have strong interoperability. Personalized care needs to be supported by the expansion of patient-specific, adaptive technologies, such as digital twins and automated treatment planners, which will also optimize clinical decision-making and resource allocation.

4. Impact of integrated ML-IoT systems in healthcare applications

By enabling strong predictive analytics, improving the dependability of medical equipment through predictive maintenance, and developing personalized and precision medicine, including pharmacogenomics to customize treatments for every patient, integrated ML and IoT systems are revolutionizing the healthcare industry and improving healthcare outcome. While data-driven orchestration of workflows, patient management, resource allocation, scheduling, and supply chains increases operational efficiency and results across care settings, continuous, remote patient monitoring enhances surgical recovery and the management of chronic diseases.^{218–222}

4.1. The role of ML-IoT in improving outcomes in healthcare sectors

Machine learning enables Predictive maintenance (PdM) frameworks to evaluate equipment conditions and predict issues for reducing downtime and expenses.^{223–227} IoT sensors provide high-frequency device and patient data into cloud or edge pipelines on the infrastructure side, where machine learning algorithms identify risks, predict failures, and initiate prompt actions. According to a study, an SVM-plus-IoT vibration framework for immunoassay analyzers reduces diagnostic/repair costs by up to 25 % with ~1-year payback, while analyses of 1350 ventilators reveal ensemble learners (e.g., Random Forest, XGBoost) achieve ~0.993 predictive accuracy and K-Nearest Neighbors approaching perfect performance.^{228,229} These findings show that combining powerful machine learning algorithms with real-time telemetry proactively surfaces anomalies and enhances safety, dependability, and cost-effectiveness.

Individualized treatment plans are created at the point of care by ML/DL models that combine lifestyle, environmental, clinical, and genomic data; IoT wearables and household appliances then produce the longitudinal signals required to sustain and improve those plans. OCR and vectorization are examples of digitization pipelines that aid in unlocking historical information, and pattern-mining of previous cases aids treatment recommendations while highlighting potential side effects to improve safety.²³⁰ Beyond broad applications in discovery (e.g., QSAR, toxicity, virtual screening), recent work like DrugGene which is combining an ANN branch for drug properties with a biologically structured visual neural network for genomics—improves accuracy and interpretability for anticancer drug-response prediction, shedding light on pathways that drive sensitivity and resistance.^{231–233} For pharmacogenomics, AI/ML combines multi-omics, clinical, and chemical descriptors to predict drug response and minimize adverse reactions.

ML-IoT technologies facilitate end-to-end remote care for chronic diseases by capturing biosignals, combining data from several sources, using machine learning for analysis, and closing the loop with cloud dashboards and warnings. Given the cardiometabolic burden—about 3.01 million deaths in 2021 were linked to excessive BMI, and around 1.9 million to cardiovascular disease—this is crucial. In order to identify early deterioration and direct self-management, wearables (such as watches, smart clothing, and patches) continuously monitor parameters like blood pressure, glucose, ECG, respiration, and exercise.^{234–238} In field deployments, scalable microservices architectures exhibit elastic, event-driven analytics by ingesting massive device streams (e.g., MQTT pipelines achieving ~16,000 payloads/s from 1000 devices) and executing real-time signal processing, anomaly detection, triage alerts, and workflow orchestration with low latency.²³⁹ While pragmatic systems incorporate emergency notifications and even live video, when necessary, continuous chest-patch monitoring integrated into hospital platforms has demonstrated feasibility and safety in the postoperative setting, improving early detection compared to intermittent measurements and enabling earlier clinical intervention.^{240–243}

Operationally, ML-IoT solutions improve experience and throughput in both home and hospital environments. Using an IoT-to-cloud-to-clinician feedback loop, physician-in-the-loop frameworks (like CURATE.AI ADAPT) learn N-of-1 dose-response profiles from telemonitored home blood pressures and provide daily, adjustable antihypertensive recommendations in treatment management, allowing for quick, iterative titration while maintaining clinician control.²⁴⁴ In order to simulate trajectories, customize tactics, and timing interventions without necessitating in-person visits, patient management is shifting toward digital twins that combine wearable streams (heart rate, SpO₂, mobility) with EHR context (diagnoses, prescriptions).²⁴⁵ ML-derived risk assessments and recommendations on cloud infrastructure help with resource allocation; 5G adoption and data management advancements further enhance timeliness and scalability for medical IoT workloads.^{246–248}

End-to-end integration of IoT telemetry with ML makes supply chains more robust and responsive. AI compares scheduled procedures with stock to anticipate shortages, suggest alternatives, and send out timely alerts. Smart dispensers also confirm patient identity, connect drugs to prescriptions, and synchronize with inventory systems. Hospitals can maintain ideal inventory levels, lower carrying costs, and guarantee that necessary drugs and supplies are available when needed via real-time sensor updates and automated ordering cycles.²⁴⁹

Integrated Machine Learning (ML) and IoT-based systems have significantly advanced healthcare by enabling predictive maintenance of medical equipment, personalized medicine, pharmacogenomics, and operational efficiency. ML models such as Random Forest, XGBoost, and SVM have improved predictive accuracy for equipment failures, minimizing downtime and ensuring uninterrupted patient care. Wearable IoT devices combined with ML support continuous patient monitoring, chronic disease management, postoperative care, and real-time telemedicine, facilitating individualized treatment plans based on physiological, lifestyle, and genetic data. Researchers should further focus on expanding patient-centered and adaptive digital twin systems, and integrating predictive analytics with hospital resource and care management to optimize treatment outcomes and healthcare delivery.

4.2. Key benefits of ML-IoT in healthcare applications

The Internet of Things (IoT) and machine learning (ML) work together to create a strong, data-driven healthcare facility that increases access, enhances clinical results, reduces costs, and facilitates prompt, individualized decisions in a variety of contexts. Continuous physiologic and operational signal acquisition is made possible by the widespread connectivity of medical sensors, wearables, and smart equipment. Machine learning models subsequently convert these high-velocity streams into useful information for remote monitoring, early diagnosis, treatment selection, and recovery support.^{250,251} Clinicians may forecast illness risk and recovery windows, customize treatment routes, and act earlier as their understanding of symptoms and trajectories develops. These abilities rely on interpreting the massive real-time datasets generated by IoT devices.^{252–254}

In order to speed up the identification of mutations, biomarkers, and therapeutic targets that guide precision diagnosis and therapy design, machine learning (ML) first identifies patterns in vast, heterogeneous biomedical corpora, such as literature, registries, hospital electronic medical records, and sensor streams.^{255–265} Second, algorithms that automate repetitive processes, minimize data entry and triage errors, and direct limited clinical care to the most vulnerable patients result in cost and resource efficiencies. CNNs, for instance, are experts at quick, precise interpretation in imaging-centric use cases, and IoT-enabled research workflows improve clinical trial operations by reducing data-based errors, unified electronic records, and real-time participant monitoring, all of which reduce costs and timelines.^{266–270} For that, Table 3 shows the three different classes of machine learning and their cost effectiveness.

By pushing analytics to the edge, systems-level architectures further reduce costs. For example, a recent three-layer remote patient monitoring design (such as sensing to fog/network then cloud/app) quadrupled the number of patients per gateway without compromising data fidelity by using adaptive sampling, filtering, and on-device preprocessing, enhancing patient engagement, safety, and physician visibility while preventing unnecessary utilization.²⁷¹

Table 3

Three different classes of machine learning and their cost effectiveness.

ML Class	Type of the Data	Types of Usage	Output Performance & Accuracy	Affected by incomplete data	Adaptable	Expense	Year & Ref.
Supervised	Labeled	Regression Classification	High	Yes	Yes, however automatic labeling of a vast amount of data is required.	Expensive	272
Semi-supervised	Both Labeled & unlabeled	Clustering Classification	Average	No	It is not suggested.	Relatively expensive	273
Unsupervised	Unlabeled	Transformations Clustering	Low	No	Yes, yet have to confirm that the output prediction is accurate.	Inexpensive	274

Table 4

Benefits of machine learning algorithms in monitoring Purpose.

Use Cases	Algorithms	Benefits	Year	Ref.
Smartphone camera-based blood pressure monitoring system	Linear Regression	Portable & non-invasive blood pressure monitoring in real-time	2020	281
Using data from present monitoring to predict future blood glucose levels	Recurrent Neural Network	Continuous monitoring and prediction of emergency diabetes patients	2020	282
System for monitoring infant health	Gradient Boosting	Detects if premature infants have bradycardia, enabling the implementation of real-time early intervention approaches	2022	283
Continuous patient monitoring to predict stroke	Naive Bayes, Random Forest	Higher performance is attained by the ensemble of multiple algorithms for continuous and real-time monitoring	2022	284
System for predicting patient falls	Naive Bayes	Conducts real time monitoring better than the other algorithms	2023	285

As ML-IoT technologies expand patient-centered services outside of clinic doors, accessibility gets better. With evidence that AI frameworks that "remember" social context (hobbies, lifestyle, and personal history) can increase patient trust, engagement, and adherence—as long as strong privacy controls are in place—decision-support systems synthesize multimodal records to suggest personalized plans that honor individual needs and preferences [272,273](#). Strong discrimination (macro-AUC $\approx$ 0.86–0.92) for outcomes like readmission, adverse events, and therapy response has been demonstrated by multimodal deep learning that combines time-series vitals, structured EHR fields, and free-text notes (e.g., clinical BERT with dense layers for tabular data and GRU for lab-s/vitals with attention) while enabling interpretable attribution for clinicians. [274–276](#)

Timely interventions rely heavily on real-time analytics and ongoing monitoring. In order to identify deterioration and promptly alert caregivers, wearables, smartphones, and bedside sensors stream signals that are preprocessed on-device or at the edge and analyzed in the cloud. This approach lowers unexpected fatalities, raises health self-awareness, and encourages preventive behaviors. Advances in foundation models increase robustness outside of controlled environments: By extracting fine-grained temporal features to calculate heart rate in real-time, Pulse-PPG, which was trained on extensive, real-world wrist PPG, outperformed models trained solely on clinical data and generalized across ambulatory circumstances. [277–279](#) By achieving F1-scores >95 % across activities (such as walking, sitting, and stair climbing), CNN-BiLSTM models with attention were able to recognize human activity and provide proactive dashboards and alerts for rehabilitation and frailty management through reliable on-device categorization and streaming. [280](#) A brief description of the reviewed papers that are relevant to the applications of machine learning methods for monitoring is provided in [Table 4](#).

When Machine Learning (ML) is employed in an Internet of Things (IoT) enabled healthcare system for decision-making, the system can make decisions regarding that particular person through collecting and evaluating information relevant to an individual's health. [286](#) Based on the data, the system will then make recommendations for the most appropriate course of action to take in order to maintain or improve that person's health. The system must learn the problem's context and the quality of the supplied data in order to perform machine learning algorithms. [287](#) These algorithms are typically not particularly robust and strong at first, but with more prior experiences gained from completing a single, repeating activity, the algorithm gets stronger. It's also important to keep in mind that when talking about utilizing computers to make decisions in the healthcare sector, the topic encompasses much more than just biomedicine, biomedical image analysis, disease detection and prediction. It also includes patient care, resource allocation, hospital volume management, medical treatment research, and public health policymaking. [288](#)

5. Key findings

After critical analysis, this review highlights several key findings in ML-IoT-based healthcare applications:

- I. In ML-IoT-based healthcare systems, supervised learning has demonstrated significant efficacy in diagnosing and predicting a range of diseases, such as cancer, cardiovascular disorders, diabetes, kidney disease, liver disease, hepatitis, thyroid conditions, and infectious diseases. The literature studies shows that ensemble models and deep learning have enhanced diagnostic and predictive performance of various diseases. Neural networks, which include CNNs and ANNs, and ensemble methods like Random Forest and XGBoost, continuously enhance latency and achieve high predictive accuracy, typically between 85 % and 95 %.
- II. Recent advancements, including two-stage synthetic tumor creation and diffusion-based anomaly detection models, exhibit improved segmentation accuracy and the capability to identify unusual or previously unobserved anomalies across many imaging modalities (MRI, CT, US). These methods emphasize that reducing the domain disparity between synthetic and real data, utilizing multi-modal generative frameworks, and integrating clinically interpretable metrics can substantially enhance model reliability, robustness, and applicability in real-world healthcare environments, providing a viable approach for more automated and scalable diagnostic systems.
- III. In chronic disease management, reinforcement learning frameworks enhance treatment recommendations, medication adherence, and home care support by integrating multimodal patient data and providing automated assistance to patients. In robotic surgery, reinforcement learning automates sub-tasks, enhances surgical precision, alleviates clinician workload, and facilitates semi-autonomous procedures while upholding safety parameters.
- IV. Furthermore, the review demonstrates that deep learning algorithms substantially surpass conventional methods in processing intricate medical data, including electronic medical records and medical imaging. Models such as LSTM, BioBERT, and transformer-based architectures showed effectiveness in extracting clinical information from noisy and inconsistent records. The integration of structured and unstructured data enhances prognostic accuracy for conditions such as glaucoma and cardiovascular risk. Explainable AI instruments like SHAP and LIME improve transparency, while cultivating clinician trust.
- V. The review demonstrates that sophisticated machine learning algorithms, such as Random Forest and XGBoost, attain elevated predictive accuracy for equipment malfunctions and patient health hazards, so ensuring safety and economic efficiency.
- VI. Integration trends demonstrate an increasing utilization of cloud-edge computing for real-time analytics, with lightweight machine learning models implemented on IoT devices enhancing latency and energy efficiency. Although numerous studies indicate favorable therapeutic outcomes, such as improved diagnostics, augmented decision support, individualized treatment, and reduced healthcare expenditures—challenges persist.

6. Discussion

The integration of ML-IoT in healthcare shows considerable potential; however, it encounters critical challenges such as interoperability issues, latency, scalability limitations, data security risks, and ethical issues. Resource-limited IoT devices managing sensitive medical information increase privacy vulnerabilities, whereas cloud and edge systems face challenges due to diverse data types. Black-box models present issues related to transparency and fairness, while variable wireless protocols may influence real-time performance. Future research directions will be examined in detail, focusing on multilayered security, enhanced edge computing, explainable AI, human-in-the-loop approaches, and emerging technologies such as quantum computing and nano-IoT.

6.1. Challenges and limitations

Healthcare systems have advanced tremendously due to the integration of ML and IoT. Internet of Things (IoT) generates a large volume of real-time data from various sensors and devices, necessitating analysis due to their complex characteristics. ML algorithms have demonstrated their crucial significance in the field of IoT. However, these technologies face some notable challenges which must be addressed when applying ML algorithms in healthcare²⁸⁹

6.1.1. Data security and patient privacy

The healthcare industry has a number of challenges when integrating machine learning (ML) with the Internet of Things (IoT), such as issues with data security, privacy, scalability, and data heterogeneity. Data security and patient privacy are major concerns for ML-IoT based healthcare. Data security and privacy issues arise because of the continuous collection, transmission, and storage of sensitive patient information. Also, IoT devices have limited resources and are vulnerable to security breaches. 72 % of hostile operations target sensitive health information, such as genetic and EMR data, hence strong encryption and privacy-preserving measures are crucial.^{290–293} Service reliability is also impacted by information security breaches, underscoring the necessity of all-encompassing protection plans for IoT-enabled devices.^{294,295}

6.1.2. Scalability and data heterogeneity

Scalability poses a considerable challenge in implementing ML-IoT solutions in the healthcare sector. The increasing prevalence of IoT devices, sensors, and patients leads to a significant rise in data generation, resulting in challenges for infrastructure management. Studies indicate that although decentralized learning approaches, such as federated learning and blockchain, effectively preserve

privacy, they face scalability challenges due to non-IID data, bandwidth fluctuations, and considerable communication overhead among distributed nodes.²⁹⁶ The management of extensive and diverse healthcare data can lead to the overloading of cloud and edge computing resources. This has a considerable impact on tasks that need low latency. Edge devices can have limited computational resources and memory which makes it difficult to run complex machine learning algorithms locally. One major limitation is that the model's performance highly depends on the quality and the size of the dataset. This shows that the limited volume of dataset is a significant challenge for healthcare.²⁹⁷

6.1.3. Challenges related to communication, network and low latency

Research and recent surveys have highlighted that communication, network reliability, and latency are one of the most crucial challenges in the ML-IoT based healthcare system. Home-based or remote patient monitoring systems face additional challenges due to heterogeneous sensor protocols (BLE, Zigbee, Sigfox, and LoRa), variable bandwidth, and device energy limitations, which can cause unpredictable delays and network failures. Limited computational resources on wearable devices restrict local ML inference, making lightweight edge AI models, compressed deep learning techniques, and job offloading essential to maintain responsiveness without overloading hardware.^{298–300}

6.1.4. Lack of interoperability and standards

Interoperability presents another critical challenge in ML-IoT healthcare ecosystems, which comprise devices and sensors from various vendors employing diverse communication protocols and data formats. Also, the absence of universal standardized frameworks obstructs seamless data integration and coordinated care across platforms. Addressing these challenges is essential for realizing the full potential of IoT in healthcare for ensuring reliable, secure, and patient-centered services.^{301,302} Manufacturers of devices frequently use diverse communication protocols and data formats, which leads to fragmented ecosystems that impede scalability and data aggregation. Efficiency, cross-device integration, and extensive healthcare deployments are still constrained in the absence of internationally recognized standards.³⁰³

6.1.5. Algorithmic and ethical challenges

Recent studies highlight that ML-IoT systems in healthcare face some algorithmic and ethical challenges, mainly due to the opaque characteristics of black-box models. Researchers emphasized the challenges faced by clinicians, patients, and stakeholders regarding the absence of transparency and interpretability in AI-driven medical decision-making. Deep models do not clarify the process by which inputs are transformed into outputs, leading to legal, ethical, and trust concerns in clinical settings.³⁰⁴ Furthermore, these models may unintentionally exhibit bias, thereby jeopardizing equitable outcomes that compromise fairness. Research has highlighted the importance of integrating Explainable AI (XAI) methods, including SHAP and LIME, to improve transparency, accountability, and ethical integrity.³⁰⁵

6.2. Future research directions

The influence of IoT-ML in Healthcare 5.0, which has lasted since 2020,³⁰⁶ is substantial and has influenced us to change our perspective on conventional treatment approaches. According to predictions for 2020, the world smart health market has reached 1436 billion USD, and between 2020 and 2027, the average growth rate may further be expanded by 16.2 %.³⁰⁷ Healthcare 5.0 focuses on customization and patient-centric models, intending to establish lifelong partnerships and improved well-being. It included the features of smart health care such as automated networks, cloud networking, artificial intelligence, big data, mobile internet, biotechnology, and IoT³⁰⁸ to advance healthcare sustainability, reduce costs, and enhance patient outcomes. Prioritizing human-centric services aims to balance cost, service quality, and benefits, addressing the satisfaction of patients, healthcare workers, and providers³⁰⁹. Fig. 7 illustrates the prioritization of Healthcare 5.0. There are several areas in this industry where making the required changes can impact healthcare in the future.

To monitor the patient's status a great number of solutions exist. However, there is still immense opportunity in the healthcare system. Thus, it is crucial to find prospective technology. Fig. 8 illustrates the various future directions for researchers.

6.2.1. Security and privacy enhancement

For ML-IoT-based healthcare systems, security and privacy continue to be major obstacles.³¹⁰ To stop eavesdropping and unwanted changes, data transfer between nodes and processing units must be secured. While maintaining the confidentiality and integrity of sensitive health records, cryptographic techniques can secure explicit data mapping and user identity.^{311,312} In addition to privacy-preserving strategies like blockchain and differential privacy, energy-efficient and lightweight machine learning (ML)-based data aggregation algorithms are required to mitigate these threats.³¹³ System safety also requires the development of tamper-resistant IoT devices and the strengthening of access control techniques.³¹⁴ It is essential to create ML-based aggregation techniques and lightweight cryptography that strike a balance between resource efficiency and security.

6.2.2. Integration of emerging technologies

Nano ML-IoT is a significant future direction for healthcare, integrating nanoscale devices, Internet of Things connectivity, and machine learning. The Internet of Nano Things (IoNT) has advanced considerably in the healthcare sector, providing expedited and more convenient methods for diagnosis, monitoring, and treatment.^{315,316} Healthcare functions like electromagnetic detection, transmission, and actuation are improved by combining opto-technology with ML-IoT-based systems.³¹⁷ By linking medical equipment and facilitating remote health data sharing, this integration makes smart healthcare possible.³¹⁸ Contactless feature extraction and enhanced data analysis are made possible by sophisticated signal processing techniques.³¹⁹ Effective data aggregation requires addressing performance degradation in dynamic and diverse situations.^{320–322} Through expert knowledge and ML-based techniques, architectures such as IoT-HDHM-ML employ networks of wearable nodes to continuously monitor numerous patient parameters in real time, hence boosting healthcare efficiency.³²³ Furthermore, Explainable AI (XAI) can provide interpretable and transparent predictions, helping clinicians trust and adopt ML-IoT systems in critical decision-making processes, while Quantum Computing offers the potential to accelerate ML model training and support complex data analysis in genomics and medical imaging, further enhancing healthcare applications. Future studies should concentrate on creating scalable, resilient opto-ML-IoT infrastructures that can function in dynamic, diverse contexts.

6.2.3. Involvement of stakeholders

The involvement of Stakeholders including professionals in the fields of engineering and health sciences as well as patients is crucial to make Healthcare 5.0 successful. Stakeholders in healthcare, also including patients, providers, and insurers, maintain a single ledger that encodes pre-authorization requirements, ensuring transparency and accountability.³²⁴ Patients must be involved in the design process and the ML-IoT technologies must be patient-centered.³²⁵ Additionally, while designing ML-IoT-based digital healthcare systems, receiving feedback from users of such systems and considering individual requirements are also essential.³²⁶ Moreover, in the healthcare system, Blockchain provides additional benefits such as real-time calculations through the automation of information distribution and it typically employs cryptographic-based techniques (such hash functions, asymmetric encryption, and digital signatures) across different system stakeholders to help maintain trust throughout interactions.³²⁷

6.2.4. Standardization of protocols

To solve interoperability issues in healthcare IoT systems, standardization and synchronization are crucial.³²⁸ Data transmission between new machine learning (ML)-based healthcare technologies and current electronic health records is made easier by ensuring that data formats are uniform and analyzable. To facilitate the incorporation of cutting-edge healthcare solutions, policymakers should give priority to data compatibility, sharing, and interoperability. The creation of protocols and universal data standards for ML-IoT healthcare systems should be the main goal of future research. Integration can be improved by looking at automated mapping and translation solutions between heterogeneous data sources. Policy-driven frameworks that direct the adoption of new technologies while preserving compliance, compatibility, and safe data transmission across various healthcare systems could also be the subject of future research.

6.2.5. Human-AI collaboration in healthcare

In the healthcare industry, human-AI collaboration refers to the use of AI systems in conjunction with people as partners to address difficult issues, such as supporting clinical decision support systems (CDSS) in the diagnosis of different diseases including cancer.^{329,330} Human-AI collaboration facilitates augmented decision making. Augmented decision-making utilizes AI insights to assist clinicians by providing evidence-based recommendations, thereby enhancing accuracy and minimizing diagnostic errors. Collaboration between doctors and AI, grounded in ethical and transparent practices, guarantees that human oversight, explainability, and fairness are integral to patient care decisions. Medical professionals can choose the best machine learning methods utilizing human-in-the-loop ensemble learning when human experience is incorporated into ML-IoT systems. This improves prediction accuracy and model performance.^{331–333} Despite obstacles including bias in CDSS, skepticism about AI, and adoption issues, this partnership can lessen the scarcity of qualified healthcare workers, enhance the quality of care, and lighten the workload of clinicians.^{334,335}

Future studies should examine methods for teaching healthcare workers how to collaborate with AI and include these techniques into healthcare education curricula. Wider adoption will be supported by research into ways to improve the usability, transparency,

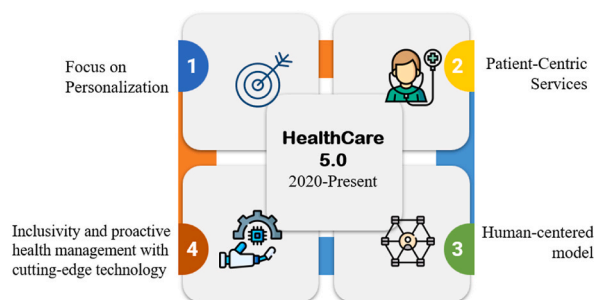


Fig. 7. Focus of healthcare 5.0.

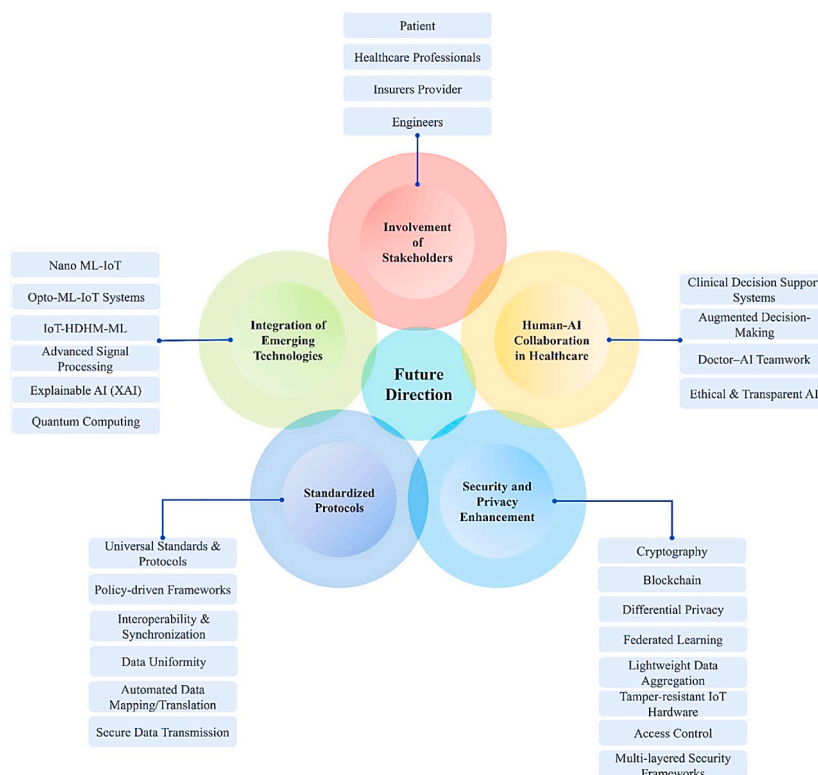


Fig. 8. Future prospects of ML and IoT in healthcare.

and trustworthiness of AI systems in therapeutic contexts.

6.2.6. Novel insights and future outlook

Future research in Healthcare 5.0 should focus on five critical domains: security, emerging technologies, stakeholder engagement, standardization, and human–AI collaboration. An enhanced multi-layered security framework is required that combines machine learning-driven data aggregation with lightweight cryptography and hybrid privacy-preserving techniques such as federated learning, blockchain, and differential privacy. Such frameworks should incorporate tamper-resistant IoT hardware and context-aware access control to improve the overall confidentiality as well as system resilience. Future research in Healthcare 5.0 should prioritize a multilayered approach that integrates security, emerging technologies, stakeholder involvement, protocol standardization, and collaboration between humans and AI. The integration of advanced technologies such as Nano ML-IoT, multimodal data fusion, scalable opto-ML-IoT infrastructures, energy-efficient wearable networks, and predictive analytics can also facilitate a dependable healthcare system. Also, active involvement of stakeholders such as patients, clinicians, engineers, and insurers is crucial. Effective patient-centered designs, transparency through shared ledgers and blockchain technology, and system modifications depend on the active participation of stakeholders. Standardizing universal data formats and protocols will also enhance safe data sharing, ensure compliance, and promote interoperability across diverse healthcare platforms. Moreover, enhancing collaboration between humans and AI requires the incorporation of AI literacy and collaborative methodologies into healthcare education. This must be accomplished for smooth integration of AI into clinical processes.

Overall, these directions highlight the novel insights and clear perspectives for future research in ML-IoT healthcare applications while addressing the unmet research needs by prioritizing safety, technological adaptability, stakeholder inclusivity, interoperability, and a focus on human–AI collaboration.

7. Conclusion

The incorporation of Machine Learning (ML) and Internet of Things (IoT) in healthcare has improved the overall accuracy of diagnosis, operational efficiency, customization of treatment and clinical decision making. Recent advancements in these technologies have enhanced patient outcomes and optimized healthcare processes, resulting in increased accuracy, efficiency and cost-effectiveness. Nevertheless, challenges such as interoperability issues, scalability limitations, data security concerns, and ethical

considerations continue to be substantial barriers. To fully realize the potential of machine learning (ML) and Internet of Things (IoT) in healthcare, it is essential to address these challenges through continuous research and technology developments. For mitigating these challenges, future healthcare research should focus on strengthening security and privacy, integrating emerging technologies, involving key stakeholders, standardizing protocols, and promoting effective human–AI collaboration. This will pave the way for a future when personalized healthcare becomes the standard. Although ML and IoT have made significant strides in healthcare, their application is not without substantial hurdles that must be carefully considered. This study conducts a critical and comprehensive analysis of the recent advancements, applications, challenges and future perspectives of machine learning and IoT in the healthcare sector. By prioritizing the mentioned essential domains, the healthcare sector will effectively utilize the advantages of machine learning (ML) and Internet of Things (IoT), ultimately resulting in enhanced, easily accessible, and tailored healthcare for people across the globe.

CRediT authorship contribution statement

Md Zonayed: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Investigation, Formal analysis, Data curation. **Rumana Tasnim:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sayma Sultana Jhara:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Formal analysis, Data curation. **Mariam Akter Mimona:** Writing – review & editing, Validation, Resources, Investigation, Formal analysis. **Molla Rashied Hussein:** Validation, Supervision, Resources, Investigation, Formal analysis. **Md Hosne Mobarak:** Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Umme Salma:** Writing – review & editing, Formal analysis.

Ethical statement

This study did not necessitate ethical approval, as the study involves no patients or human subjects for experiments, no clinical drug trial, poses no possible harm or dangers to participants and does not contain any confidential or sensitive information.

Declaration of competing interest

The authors declare the following financial interests (e.g., any funding for the research project)/personal relationships (e.g., the author is an employee of a profitable company) which may be considered as potential competing interests: MD Zonayed, Rumana, Tasnim, Sayma Sultana Jhara, Mariam Akter Mimona, Molla Rashied Hussein, Md Hosne Mobarak, Umme Salma reports administrative support, equipment, or supplies, statistical analysis, and writing assistance were provided by Dreamers Lab. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviation

ML	Machine Learning
IoT	Internet of Things
AI	Artificial Intelligence
CGM	Continuous Glucose Monitoring
T1D	Type 1 Diabetes
DL	Deep Learning
ANN	Artificial Neural Networks
RMSE	Root Mean Square Error
NIRS	Near-Infrared Spectroscopy
NIR	Near-Infrared
WSN	Wireless Sensor Network
HER	Electronic Health Records
ECG	Electrocardiogram

KNN	K-Nearest Neighbors
SVM	Support Vector Machine
NB	Naive Bayes
DT	Decision Trees
RFC	Random Forest Classifier
CAD	Computer-Aided Diagnostic
DICOM	Digital Imaging and Communications in Medicine
LDR	Light Dependent Resistor
CNN	Convolutional Neural Networks
PPG	Photoplethysmography

Data availability

No data was used for the research described in the article. The authors confirm that the data supporting the findings of this study are available within the article.

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